Customer Churn Analysis & Prediction

Executive Dashboard Report

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1. Executive Summary

This report outlines the findings from the customer churn analysis and the subsequent predictive modeling. The dataset contains a total cohort of **N = 7,043** customers.

- **Overall Retention Rate:** 73.5% (5,174 customers)
- **Overall Churn Rate:** 26.5% (1,869 customers)

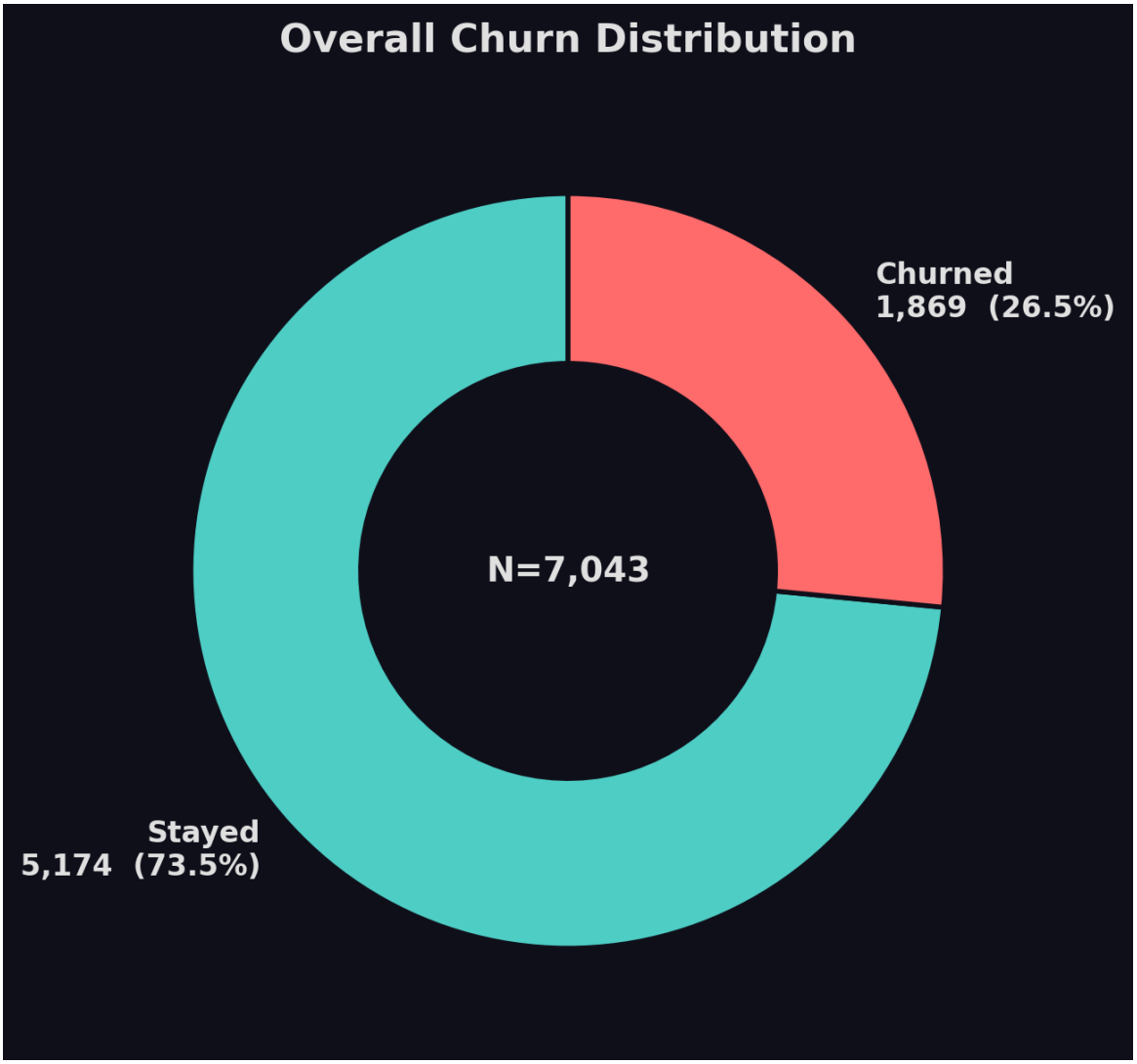

Figure 1: Overall Churn Distribution

2. Key Risk Factors

A. Contract Type

Contract length is the most significant indicator of churn risk. Customers on month-to-month plans are at an extreme risk of churning compared to those locked into annual or biennial contracts.

Contract Type	Historical Churn Rate
Month-to-month	**42.7%**
One year	11.3%
Two year	2.8%

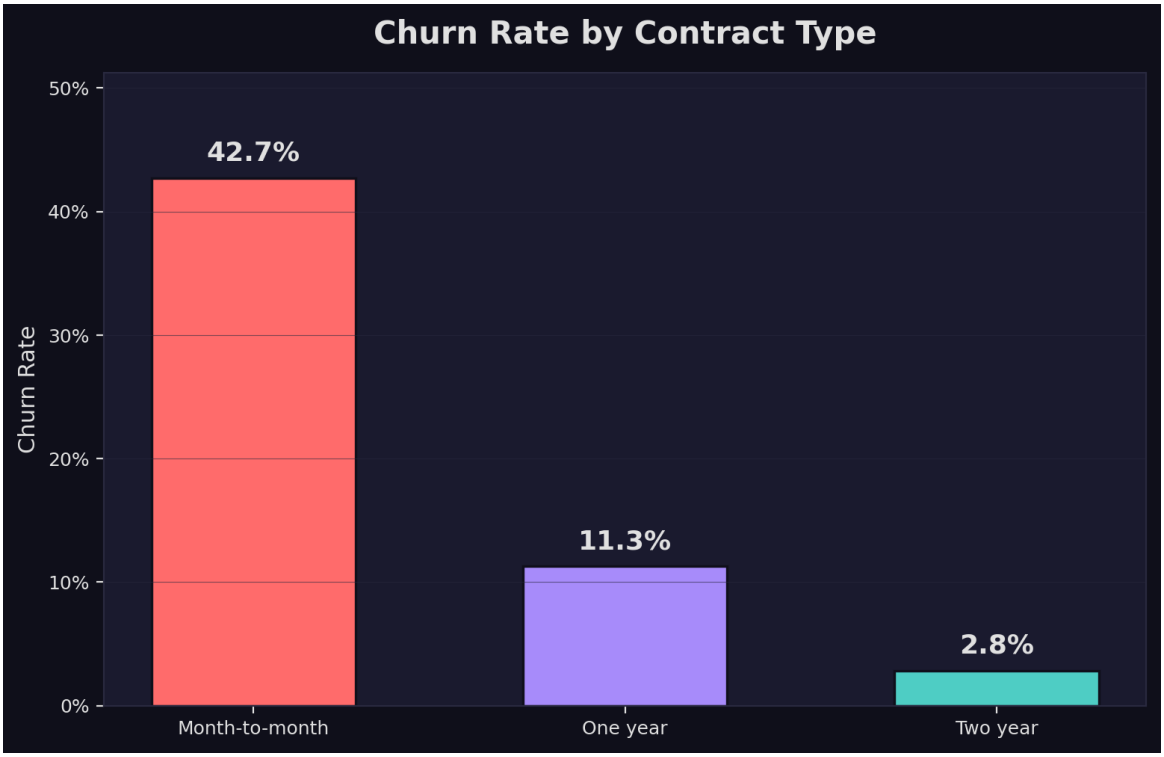

Figure 2: Churn Rate by Contract Type

B. Payment Methods

The method by which a customer pays their bill also correlates strongly with their likelihood to leave the service. Electronic checks represent a critical friction point.

Payment Method	Historical Churn Rate
Electronic check	**45.3%**
Mailed check	19.1%
Bank transfer (automatic)	16.7%
Credit card (automatic)	15.2%

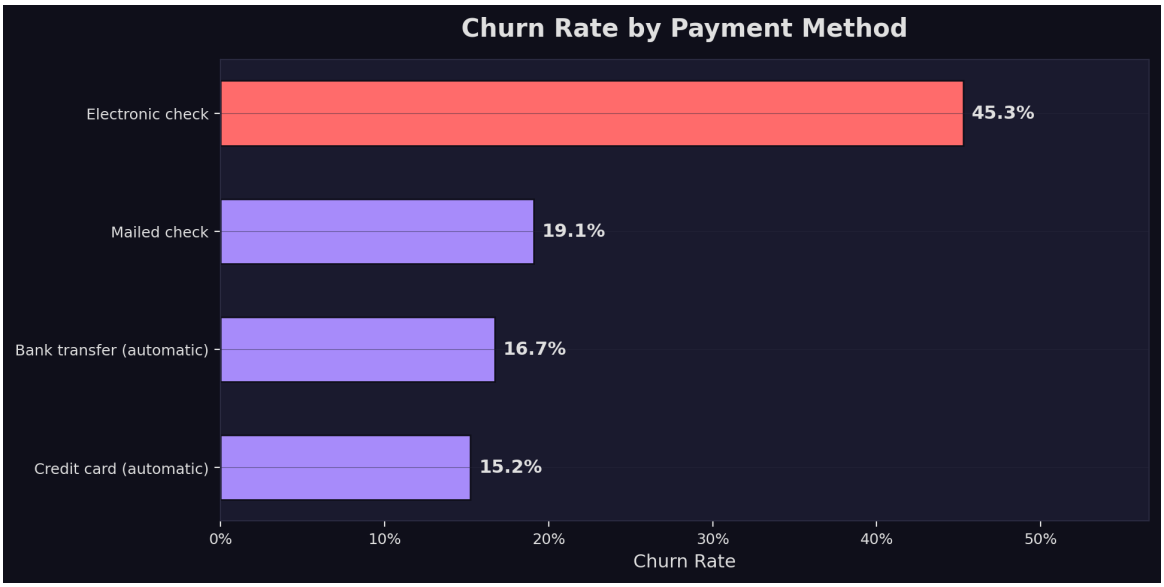

Figure 3: Churn Rate by Payment Method

3. Financial Impact & Tenure

An analysis of monthly charges reveals that customers who churn tend to have higher monthly bills.

- **Retained Customers:** Median Monthly Charge = $64.40
- **Churned Customers:** Median Monthly Charge = $79.70

Furthermore, the tenure histogram indicates that the highest volume of churn occurs within the first 1 to 5 months of service, emphasizing the need for robust onboarding and early intervention strategies.

4. Predictive Modeling Performance

Three machine learning models were evaluated to predict future churn. The models were trained using SMOTE to handle class imbalance.

Model Algorithm	CV Accuracy (SMOTE)	Test Accuracy
Random Forest	**0.840**	**0.764**
XGBoost	0.819	0.741
Decision Tree	0.787	0.718

Champion Model: The Random Forest classifier was selected for deployment due to its superior cross-validation and test accuracy.

Top 5 Predictive Features

According to the Random Forest feature importance matrix, the top drivers of churn predictions are:

1. **Total Charges** (Importance: 0.135)
2. **Monthly Charges** (Importance: 0.135)
3. **Contract Type** (Importance: 0.124)

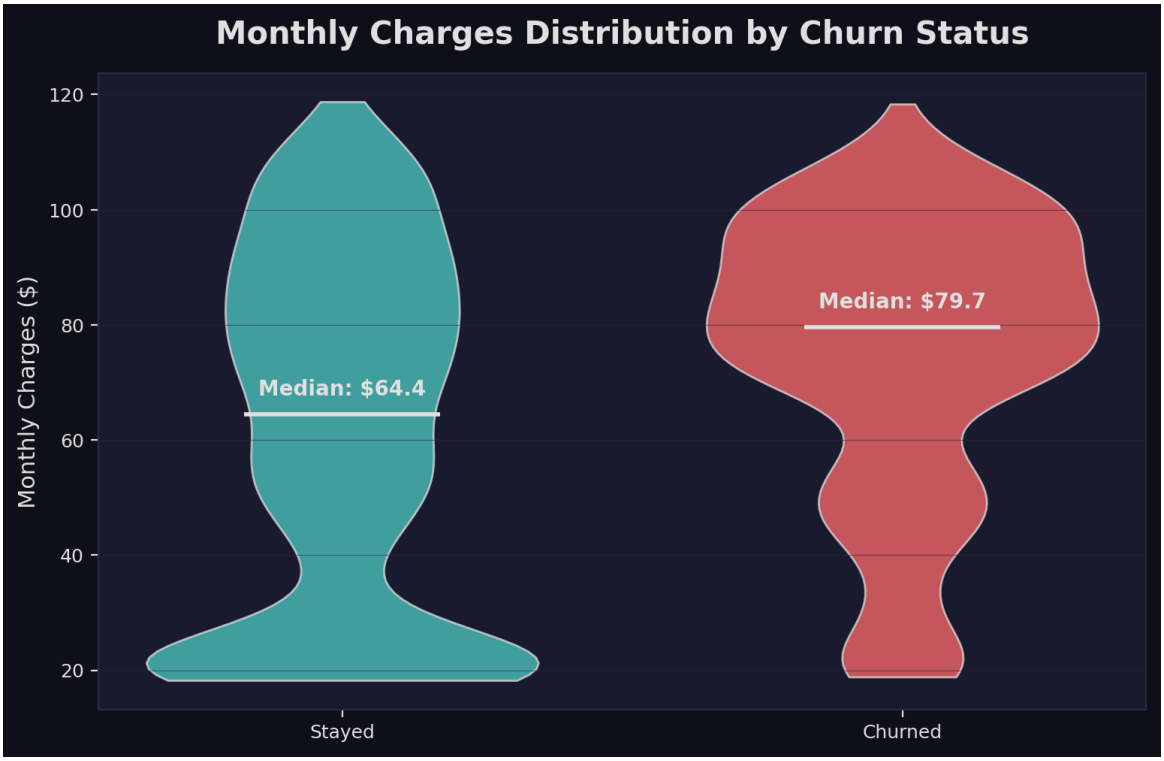

Figure 4: Monthly Charges Distribution by Churn Status

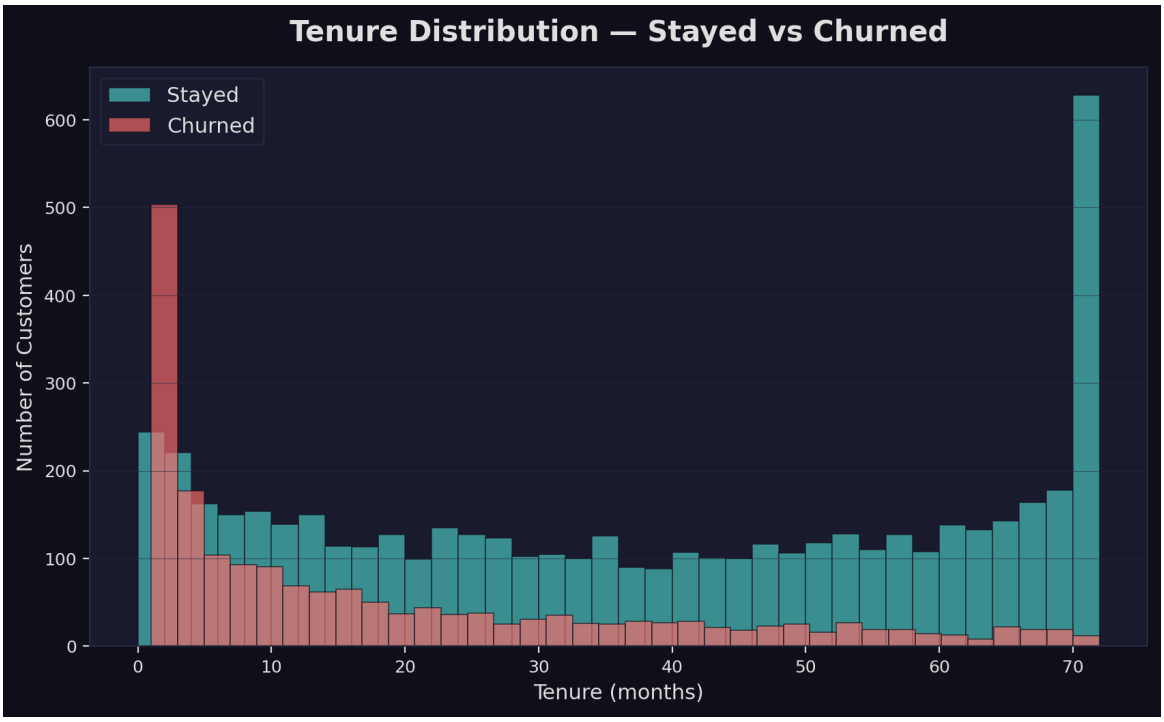

Figure 5: Tenure Distribution — Stayed vs Churned

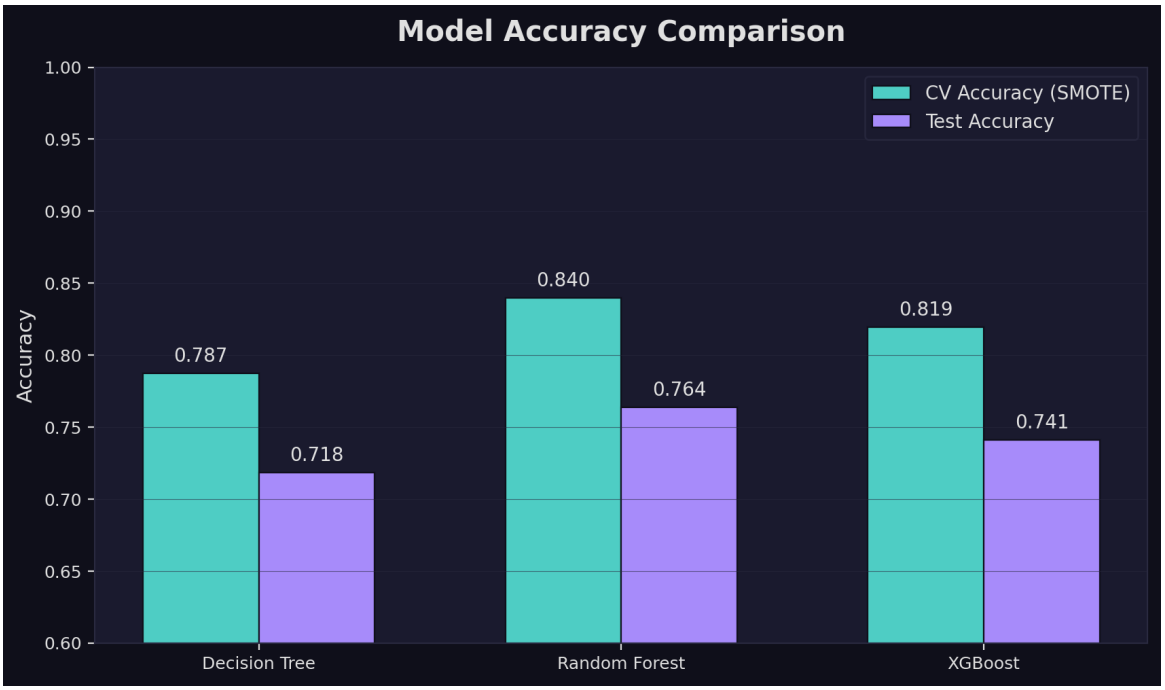

Figure 6: Model Accuracy Comparison

4. **Tenure** (Importance: 0.114)
5. **Online Security Enrollment** (Importance: 0.099)

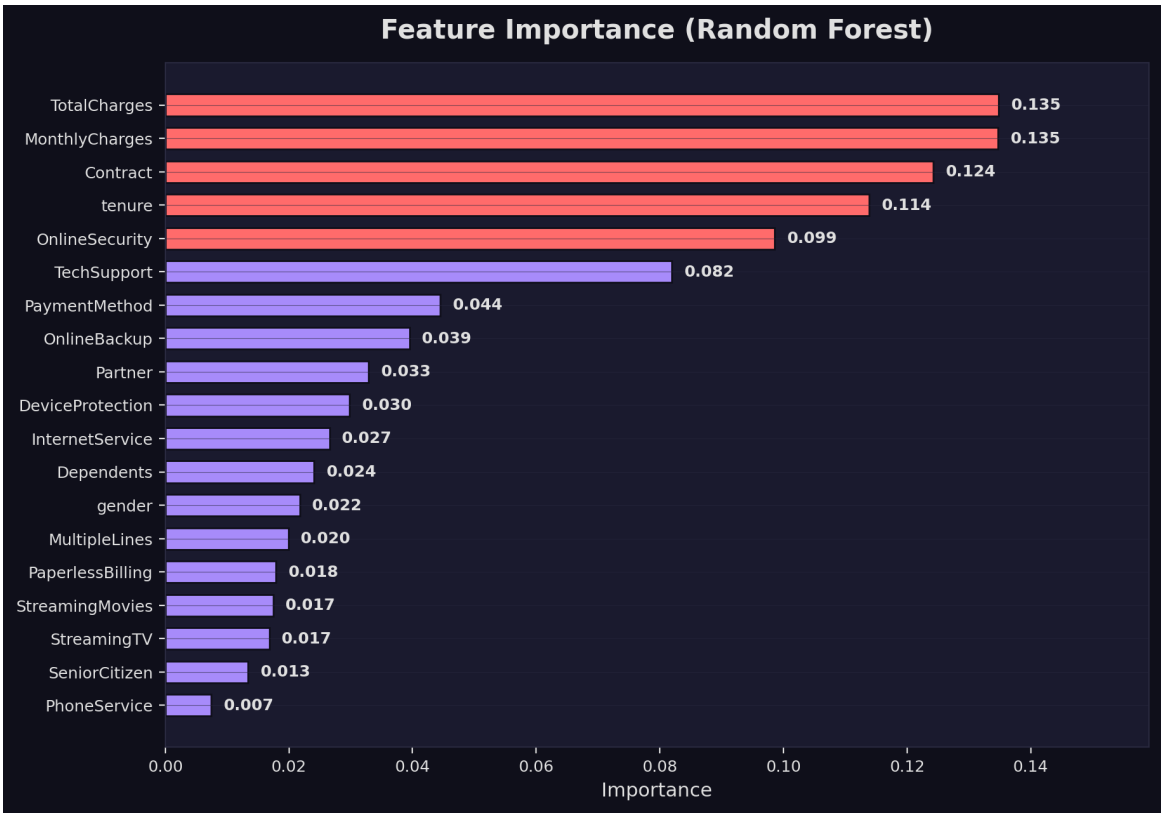

Figure 7: Feature Importance (Random Forest)